

KAND CA at NADI 2026 Shared Task: Multi-Stage Fine-Tuning for Spoken Arabic Dialects

Ahmed Wasfy
KAND CA
ahmed.wasfy@kand.ca

Nourhan Anber
KAND CA
nourhan.anber@kand.ca

Abstract

We describe the KAND CA systems for all five subtasks of the NADI 2026 shared task on multidialectal Arabic speech processing. For the three ASR subtasks, a single family of *staged, fully fine-tuned* Whisper large-v3 models — broad dialectal adaptation, then only task-matched gold data — is combined with task-specific levers: an mT5 spelling-correction post-editor trained on the task references (Subtask 1.1, final country-average WER **0.5111**), chained continue-fine-tuning through the Subtask 1.1 data before the Mixed-ASR dev split (Subtask 1.2, WER **0.48**), and a fresh two-epoch schedule over a 46K-clip external gold Tunisian–French corpus with an accent-only French post-pass (Subtask 1.3, WER **15.21%**). For spoken dialect identification (Task 2), where in-domain accuracy *anti*-predicted blind-test accuracy, our system fuses an architecturally distinct FastConformer-encoder classifier, a decorrelated enhanced-audio view, and an ASR-transcript text route, reaching **0.5114** accuracy. For dialectal TTS (Subtask 3) we submitted 4,001 voice-cloned clips across ten dialects (UTMOS 2.72). We also report extensive negative results and release our models, correctors, and code publicly.

1 Introduction

Arabic speech processing is complicated by diglossia and by wide phonological, lexical, and orthographic variation across national dialects: systems trained predominantly on Modern Standard Arabic (MSA) degrade sharply on dialectal speech. The NADI 2026 shared task (Sullivan et al., 2026), the second multidialectal Arabic *speech* edition of the NADI series (Talafha et al., 2025), targets this failure mode across five subtasks: robust multi-dialect ASR (1.1), mixed-dialect ASR (1.2), Tunisian code-switched ASR (1.3), 20-way spoken dialect identification (Task 2), and dialectal TTS (Subtask 3).

Our strategy has two pillars. For ASR, we *stage* the adaptation of Whisper large-v3 (Radford et al.,

2023): a broad dialectal-mixture full fine-tune first, then successive gentle continue-fine-tunes on *task-matched gold data only*, with task-specific final levers (learned spelling correction, chained fine-tuning, disciplined scheduling). For dialect identification, where the blind test is severely out-of-domain, we accumulate decorrelated *information routes* at conservative weights — an architecturally alien encoder, an enhanced-audio view, then a text route — rather than maximizing in-domain fit.

Our final systems scored 0.5111 country-average WER (1.1), 0.48 WER (1.2), 15.21% WER (1.3), and 0.5114 accuracy (Task 2). Beyond the recipes, we contribute a map of what does *not* work: dev-set signals that invert on the blind test, silver labels, specialist routing, seven collapsing dialect-ID backbones, test-side enhancement, decoding search, and a fluent LLM post-editor costing 5.9 WER points. Our models, correctors, probe sets, and training/inference scripts are public.¹

2 Background

Table 1 summarizes the five subtasks (we follow the official subtask numbering). Subtask 1.1 covers Algeria, Egypt, Jordan, Mauritania, Morocco, Palestine, UAE, and Yemen; submissions are one text file per dialect in dataset row order. Subtask 1.2 has *no train split* — its 3,152-clip dev set is, to our knowledge, the only labeled mixed-dialect Arabic ASR data available anywhere; clips run up to ~4 minutes. Subtask 1.3 uses the TEDx-domain NADI_TUN_ASR_2026 corpus (Bougares, 2026), in which 73% of transcripts code-switch into French or English; its scorer keeps Latin accents (French *château* $\neq$ *chateau*) and does not normalize Arabic orthographic variants, so the gold *convention* is scored, not just content. Task 2 uses the ADI-20 label set (MSA plus 19 country dialects), extending

¹<https://huggingface.co/oddadmix>; TTS code: <https://github.com/Oddadmix/Lahgtna-OmniVoice>.

Task	Input → Output	Official data	Metric	Blind test
1.1 Robust ASR	8-dialect speech → text	23.6K train / 10.8K val	country-avg WER, CER	8 × 500 clips
1.2 Mixed ASR	mixed-dialect speech → text	3,152 dev (no train)	WER	427 clips
1.3 Code-switched ASR	Tunisian+FR/EN → text	24.9K train / 731 val	WER	2,123 clips
Task 2 Dialect ID	speech → 1-of-20 label	ADI-20	accuracy, C_{avg}	878 clips
Subtask 3 TTS	text + dialect → speech	— (test prompts only)	UTMOS, NISQA, WER	4,001 prompts

Table 1: The five NADI 2026 subtasks we participated in. See [Sullivan et al. \(2026\)](#) for full task details.

ADI-17 ([Shon et al., 2020](#)), and scores accuracy plus the NIST LRE detection cost over submitted logits.

Input/output examples. ASR maps a 16 kHz clip to an Arabic-script transcript (romanized here); an illustrative 1.3-style reference mixes a Tunisian matrix with French islands — “*bech naàmlou le suivi fi el-projet*” — the gold attaching the article *el-* to Latin tokens. Task 2: clip → one of 20 labels (e.g. Morocco) + logits; Subtask 3: (dialect, text) → 24 kHz waveform.

Whisper-scale models provide strong Arabic substrates but skew toward MSA ([Radford et al., 2023](#)); Casablanca ([Talafha et al., 2024](#)) showed the value of gold dialectal supervision for exactly the 1.1 dialect set.

3 System Overview

3.1 Shared ASR backbone

All three ASR systems start from whisper-large-v3-arabic-dialectal-v4, our full fine-tune of Whisper large-v3 (1.5B) on a broad dialect-balanced mixture: our public lahghtna corpus ([KAND CA, 2026a](#)) (54.6K YouTube-sourced clips / 267 h, balanced over 13 dialects), Common Voice 18 Arabic ([Ardila et al., 2020](#)), FLEURS ([Conneau et al., 2023](#)), and MASC ([Al-Fetyani et al., 2021](#)). This shifts the model’s prior away from MSA before any task data is seen (contamination audit: Appendix B). All fine-tunes are *full* fine-tunes (cross-entropy, bf16, gradient checkpointing); continue-fine-tune rounds use $\text{lr } 5 \times 10^{-6}$ with a paged 8-bit AdamW ([Dettmers et al., 2022](#)) and best-on-holdout checkpoint selection. Long-form decoding uses a chunked pipeline (30 s chunks, 5 s stride) with a post-hoc *repeat-collapse* pass that caps immediately repeated 1–4-grams at two copies; this recipe is essential (naïve batched generation inflates 1.1 dev WER 0.579 → ≈ 0.80).

3.2 Subtask 1.1: staged fine-tuning + learned spelling correction

The acoustic model is staged through: (i) the dialectal base above; (ii) full fine-tune on the concatenated task material (nadi2026-full-v4; blind 0.57); (iii) continue-fine-tunes folding in Casablanca gold (its released val+test data — it ships no train split — as permitted external resources; blind 0.56), targeted Algerian conversational data (kahwa-postcast, ~ 7.2 K French code-switched clips; blind 0.5576), the official train set with Casablanca replay (blind 0.52), and finally the *full* 13,637-clip Casablanca corpus (in-domain holdout 0.4144 → 0.4021; blind 0.52).

The final gain comes in *text space*: mT5-large ([Xue et al., 2021](#)) spelling correctors fine-tuned on synthetic corruptions of *the task references themselves*, applied to every hypothesis. Eight per-dialect correctors yield **0.5111** (from 0.52); a single multidialect corrector ties within leaderboard resolution (reported 0.51) while being $8\times$ simpler. Trained on the task references, the correctors learn the gold orthographic convention (hamza seats, alif forms, ta-marbuta) — the property that makes or breaks ASR post-editing (§B).

3.3 Subtask 1.2: chained continue-fine-tuning

The winning system *chains* adaptation: the model must first be good at 1.1, and only then see the Mixed-ASR data. With no train split shipped, we adapt on the dev split itself (the released labeled data): continue-fine-tuning our best 1.1 model on it (3,002 train / 150 holdout) plus replay (1,500 official-1.1 + 1,000 Casablanca clips) against forgetting. This scores **0.48** (base zero-shot: 0.51); notably, the same 1.1-helpful data applied *directly* to 1.2 regressed to 0.52. Checkpoint selection transferred: the best-on-holdout checkpoint beat the final one on the blind test (0.48 vs. 0.49). One operational hazard dwarfs modeling: the headerless prediction file’s line order is *split-specific* (dev sorted-by-ID; test in parquet order); the wrong order scores ≈ 1.4 .

3.4 Subtask 1.3: external gold + disciplined scheduling

From the dialectal base (zero-shot 87.6% WER on the public test split), in-task full fine-tuning with SpecAugment (Park et al., 2019) ($p_{\text{time}} = p_{\text{feat}} = 0.05$) reaches 20.5. The decisive ingredient is tunisian-asr-cleaned (FARUKx-AUTO, 2026), 46K external gold Tunisian–French code-switched clips: mixed with a TEDx replay (70,595 clips total) it yields 15.31 after one epoch. Scheduling then matters more than data: warm-continuing one more epoch *regressed* (15.4), whereas re-running *fresh* with a single 2-epoch cosine schedule won (faruk-v7, 15.22); four epochs overfit. A final post-pass restores French accents via a spellchecker with an accent-only filter (unaccent(fix)==token, provably unable to change letters); it touched 5 tokens in 2,123 clips for the final 15.21.

3.5 Task 2: decorrelated routes for OOD dialect ID

Our development overturned its own assumptions at each phase. *Phase 1*: a four-way softmax ensemble of MMS (Pratap et al., 2024), w2v-BERT 2.0 (Chung et al., 2021; Seamless Communication team, 2023), Whisper-encoder, and ECAPA-TDNN (Desplanques et al., 2020; Valk and Alumäe, 2021) classifiers reached 95.21% on a 10,806-clip holdout — and collapsed to 0.22–0.33 on the blind test, in *reverse* order of holdout strength. *Phase 2*: a single attention-pooled classifier on the FastConformer (Rekesh et al., 2023) encoder of cohere-transcribe-arabic (Cohere Labs, 2026) — weaker in-domain (80.07%) — scored 0.48; fusing its raw-audio posteriors with those on audio processed by LavaSR (YatharthS, 2025), a reconstruction-style enhancement model — a decorrelated test-time view (61% agreement) — reached 0.50. Seven alternative backbones all collapsed (0.05–0.33), and a higher-capacity head on the winning encoder raised in-domain accuracy to 84.26% while *cutting* blind accuracy to 0.38 — the in-domain/OOD inversion reproduced within a single encoder. *Phase 3*: we add a **text route** — transcribe each test clip with our 1.1 ASR system, classify the transcript with a BERT (Devlin et al., 2019) dialect classifier trained on raw dialectal text (15 labels, of which 14 map into ADI-20; unmapped classes receive no text mass), and fuse:

$$\mathbf{p} \propto \frac{1}{2} \mathbf{p}_{\text{raw}} + \frac{1}{2} \mathbf{p}_{\text{lava}} + w \mathbf{p}_{\text{text}}, \quad (1)$$

with $\mathbf{p}_{\text{raw}}, \mathbf{p}_{\text{lava}}$ the classifier’s posteriors on raw/enhanced audio and $\mathbf{p}_{\text{text}}$ the mapped text posterior. Though the text argmax agrees with the audio on only 16.7% of covered clips, $w=0.30$ reaches 0.5114, with a sharply peaked response (§5). Counterintuitively, feeding the text route *better* (corrector-post-edited) transcripts scored 0.509: the text classifier was trained on raw dialectal text, and distribution match beats transcript quality.

3.6 Subtask 3: dialectal voice-cloning TTS

We submitted 4,001 clips across the ten test dialects from lahgtina-omnivoice-v2 (KAND CA, 2026b), our open 612M-parameter voice-cloning dialectal Arabic TTS model (24 kHz). One reference speaker per dialect drives all clips of that dialect; six of the thirteen dialect references are human-curated (verified audio with accurate transcripts), the rest use demo-space references with automatic transcription. Two unsupported test dialects fall back to their nearest supported neighbor (UAE→Saudi, Jordan→Palestinian). Reference quality matters: a faulty auto-transcribed reference inflated one dialect’s outputs to 12.8 s per clip, vs. 2.2 s with a corrected reference. Official automatic evaluation: aggregate UTMOS 2.72 (Saeki et al., 2022), NISQA-MOS 3.57 (Mittag et al., 2021), WER 0.32 / CER 0.11 via Omnilingual ASR 7B (Meta AI, 2025); per-dialect: Table 6.

4 Experimental Setup

All ASR/DID models were trained on a single 32 GB GPU with PyTorch and Transformers (Wolf et al., 2020) (versions and URLs in Appendix A, Table 3); full-parameter large-v3 requires the paged 8-bit optimizer to fit. In-task fine-tunes use lr 1×10^{-5} ; continue-fine-tunes lr 5×10^{-6} with load_best_model_at_end on an in-domain holdout. mT5 correctors are trained in fp32 (mT5 is numerically unstable in bf16) with an instruction prefix, decoding greedy/beam-4. Dialect-ID classifiers use two-phase training (frozen-encoder head warm-up, then full fine-tune) with class-weighted cross-entropy on the 67K-clip curated ADI-20 subset. We used official scorers and split conventions throughout (jiwer served only for local proxy evaluation); all blind-test numbers below are official leaderboard evaluations, with post-submission results marked. Each progression row in Table 2 was one official blind submission, plus a few additional variants per task. Full hyperparameters are

Subtask 1.1 system	WER	Subtask 1.3 system	WER%
Stage-2 task FT (nadi2026-full-v4)	0.57	in-task full FT + SpecAugment	20.5
+ Casablanca continue-FT	0.56	+ external gold, 1 ep warm	15.31
+ Algerian conversational data	0.5576	fresh 2-epoch schedule (faruk-v7)	15.22
+ official train data (+ replay)	0.52	+ French accent restoration	15.21
+ full Casablanca (submitted acoustic)	0.52	+ Qwen LoRA corrector (post-sub.)	21.15
+ 8 per-dialect mT5 correctors	0.5111		
+ single multidialect corrector	0.51		
Subtask 1.2 system	WER	Task 2 system	Acc
base, zero-shot + decode recipe	0.51	4-way in-domain ensemble (95.21% holdout)	0.33
1.1-helpful continue-FTs, direct	0.52	FastConformer classifier, solo	0.48
chained Mixed-ASR FT (best ckpt)	0.48	+ LavaSR test-time view	0.50
same, final checkpoint	0.49	+ text route, $w=0.15$	0.5057
+ SADA gold / + correctors	0.48	+ text route, $w=0.30$ (submitted)	0.5114
		+ text route, $w=0.50$	0.50

Table 2: Official blind-test progression for all four scored subtasks. All rows are official evaluations except where marked post-submission. WER is shown as a fraction for 1.1/1.2 and as percent for 1.3, following each leaderboard’s convention; accuracy as a fraction.

in Appendix A.

5 Results

Table 2 gives the blind-test progression of every submitted system. Caveats: the blind sets are small (427–4,000 clips), so third-decimal differences span a handful of clips — we treat deltas ≤ 0.01 as ties; the Task 2 leaderboard ranked accuracy and did not expose per-submission C_{avg} (in-domain ensemble: 0.0170).

Only task-matched gold transfers. Across 1.1 and 1.2, every blind-test win came from official task data or organizer-released gold (Casablanca); silver labels, off-distribution “new dialect data,” and adjacent-dialect gold (SADA Khaliqi, [SDAIA, 2022](#); holdout win, blind tie) improved proxies but not the test. The dev signal *inverted* three times on 1.1: both decisive interventions (Casablanca, the correctors) looked neutral-to-worse locally yet won on the blind test.

Error analysis: convention, not content. The dominant recoverable error class in dialectal Arabic ASR is orthographic convention. On 1.1, the correctors’ wins come from normalizing hamza seats, alif variants, and ta-marbuta toward the reference convention (809/4,000 hypotheses changed); on 1.3, the same lever *reverses*: a Qwen3.5-4B LoRA corrector ([Qwen Team, 2026b](#); [Hu et al., 2022](#)) whose edits looked excellent on inspection — completing truncated French (*communaut* $\rightarrow$ *communauté*), fixing agreement (*ignorer* $\rightarrow$ *ignore*) — cost 5.9 WER points (15.21 $\rightarrow$ 21.15), because a model fine-tuned on the task gold already emits the reference

convention, so a differently-conventioned corrector can only move hypotheses away from it. The second error class is long-form repetition loops: post-hoc repeat-collapse recovers ≈ 2.7 WER points on 1.2, while n-gram blocking *hurts* (it deletes legitimate Arabic reduplication). On Task 2 the audio and text routes agree on only 16.7% of covered clips yet fusion gains a point — their errors are largely disjoint; decorrelation, not accuracy, pays.

In-domain accuracy anti-predicts OOD. On Task 2 the holdout ranking of systems is the reverse of the blind-test ranking (95.21% $\rightarrow$ 0.33 vs. 80.07% $\rightarrow$ 0.48), across encoders, data scales (a 160K cross-source model: holdout 94.54%, blind 0.22), and head capacities. Audio “cleaning” consistently hurts dialect ID — channel and prosodic texture carry dialect signal — though the same enhancer helps as a *decorrelated second view*.

6 Conclusion

We presented a family of staged, fully fine-tuned Whisper systems for three Arabic ASR subtasks (0.5111, 0.48, 15.21% WER), a decorrelated multi-route fusion for out-of-domain dialect identification (0.5114 accuracy), and a voice-cloned dialectal TTS submission (UTMOS 2.72). The lessons: feed staged adaptation only task-matched gold; treat output order and orthographic convention as first-class components; distrust local proxies; under domain shift, add routes rather than fit. The open problems are data problems — Mixed ASR is bound by its single 3,152-clip labeled set; Mauritanian ASR stayed near 0.78 WER throughout — and call for collection, not adaptation.

Limitations

Our systems are tuned to the NADI 2026 orthographic and scoring conventions; the correctors in particular are convention-bound by design and will not transfer to differently-transcribed references. Several final choices (corrector on/off, fusion weight w) were selected on the leaderboard because local dev signals were demonstrably unreliable — at the margin the blind test served as our development set (submission counts: Table 2). Compute was one 32 GB GPU, so ablations were run once rather than over multiple seeds. The YouTube-sourced lahgtna corpus is used under research terms. The TTS fallbacks for unsupported dialects cannot capture those dialects’ actual phonology.

Acknowledgments

We thank the NADI 2026 organizers and the maintainers of the open corpora and models our systems build on.

References

- Mohammad Al-Fetyani, Muhammad Al-Barham, Gheith Abandah, Adham Alsharkawi, and Maha Dawas. 2021. *MASC: Massive Arabic speech corpus*. *IEEE Dataport*.
- Rosana Ardila, Megan Branson, Kelly Davis, Michael Kohler, Josh Meyer, Michael Henretty, Reuben Morais, Lindsay Saunders, Francis Tyers, and Gregor Weber. 2020. *Common Voice: A massively-multilingual speech corpus*. In *Proceedings of the Twelfth Language Resources and Evaluation Conference (LREC)*.
- Fethi Bougares. 2026. *NADI_TUN_ASR_2026: A Tunisian code-switched ASR dataset*. Hugging Face dataset.
- Yu-An Chung, Yu Zhang, Wei Han, Chung-Cheng Chiu, James Qin, Ruoming Pang, and Yonghui Wu. 2021. *w2v-BERT: Combining contrastive learning and masked language modeling for self-supervised speech pre-training*. In *IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*.
- Cohere Labs. 2026. *cohere-transcribe-arabic-07-2026*. Hugging Face model.
- Alexis Conneau, Min Ma, Simran Khanuja, Yu Zhang, Vera Axelrod, Siddharth Dalmia, Jason Riesa, Clara Rivera, and Ankur Bapna. 2023. *FLEURS: Few-shot learning evaluation of universal representations of speech*. In *2022 IEEE Spoken Language Technology Workshop (SLT)*.
- Brecht Desplanques, Jenthe Thienpondt, and Kris Demuynck. 2020. *ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification*. In *Proceedings of Interspeech 2020*.
- Tim Dettmers, Mike Lewis, Sam Shleifer, and Luke Zettlemoyer. 2022. *8-bit optimizers via block-wise quantization*. In *International Conference on Learning Representations (ICLR)*.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. *BERT: Pre-training of deep bidirectional transformers for language understanding*. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*.
- FARUKxAUTO. 2026. *tunisian-asr-cleaned: A gold Tunisian–French code-switched ASR corpus*. Hugging Face dataset.
- Google DeepMind. 2026. *Gemini 3.5 flash*. Multi-modal large language model.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, and Weizhu Chen. 2022. *LoRA: Low-rank adaptation of large language models*. In *International Conference on Learning Representations (ICLR)*.
- KAND CA. 2026a. *Lahgtna: A dialect-balanced multi-dialect Arabic ASR corpus*. Hugging Face dataset.
- KAND CA. 2026b. *Lahgtna-OmniVoice: Dialectal Arabic voice-cloning text-to-speech*. Hugging Face model.
- Meta AI. 2025. *Omnilingual ASR: Open-source speech recognition for 1,600+ languages*. Model release.
- Gabriel Mittag, Babak Naderi, Assmaa Chehadi, and Sebastian Möller. 2021. *NISQA: A deep CNN-self-attention model for multidimensional speech quality prediction with crowdsourced datasets*. In *Proceedings of Interspeech 2021*.
- El Moatez Billah Nagoudi, AbdelRahim Elmadany, and Muhammad Abdul-Mageed. 2022. *AraT5: Text-to-text transformers for Arabic language generation*. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*.
- Daniel S. Park, William Chan, Yu Zhang, Chung-Cheng Chiu, Barret Zoph, Ekin D. Cubuk, and Quoc V. Le. 2019. *SpecAugment: A simple data augmentation method for automatic speech recognition*. In *Proceedings of Interspeech 2019*.
- Vineel Pratap, Andros Tjandra, Bowen Shi, Paden Tomasello, Arun Babu, Sayani Kundu, Ali Elkahky, Zhaoheng Ni, Apoorv Vyas, Maryam Fazel-Zarandi, Alexei Baevski, Yossi Adi, Xiaohui Zhang, Wei-Ning Hsu, Alexis Conneau, and Michael Auli. 2024. *Scaling speech technology to 1,000+ languages*. *Journal of Machine Learning Research*, 25(97).

- Qwen Team. 2026a. [Qwen3-ASR: Multilingual speech recognition models](#). Hugging Face model collection.
- Qwen Team. 2026b. [Qwen3.5: Large language models](#). Hugging Face model collection.
- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. [Robust speech recognition via large-scale weak supervision](#). In *Proceedings of the 40th International Conference on Machine Learning (ICML)*.
- Dima Rekesh, Nithin Rao Koluguri, Samuel Kriman, Somshubra Majumdar, Vahid Noroozi, He Huang, Oleksii Hrinchuk, Krishna Puvvada, Ankur Kumar, Jagadeesh Balam, and Boris Ginsburg. 2023. [Fast Conformer with linearly scalable attention for efficient speech recognition](#). In *IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*.
- Takaaki Saeki, Detai Xin, Wataru Nakata, Tomoki Koriyama, Shinnosuke Takamichi, and Hiroshi Saruwatari. 2022. [UTMOS: UTokyo-SaruLab system for VoiceMOS challenge 2022](#). In *Proceedings of Interspeech 2022*.
- SDAIA. 2022. [SADA: Saudi audio dataset for Arabic \(Khaliiji subset\)](#). Hugging Face dataset (community subset).
- Seamless Communication team. 2023. [SeamlessM4T: Massively multilingual and multimodal machine translation](#). *arXiv preprint arXiv:2308.11596*.
- Suwon Shon, Ahmed Ali, Younes Samih, Hamdy Mubarak, and James Glass. 2020. [ADI17: A fine-grained Arabic dialect identification dataset](#). In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*.
- Peter Sullivan, Bashar Talafha, Ahmed Ashraf, Fethi Bougares, Haroun Elleuch, Chiyu Zhang, AbdelRahim Elmadany, Youssef Mohamed, Salima Mdhafar, Yannick Estève, Mohamed Elhoseiny, Hamzah Luqman, Nizar Habash, and Muhammad Abdul-Mageed. 2026. [NADI-2026: The Second Multidialectal Arabic Speech Processing Shared Task](#). In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.
- Bashar Talafha, Karima Kadaoui, Samar Mohamed Magdy, Mariem Habiboullah, Chafei Mohamed Chafei, Ahmed Oumar El-Shangiti, Hiba Zayed, Mohamedou Cheikh Tourad, Rahaf Alhamouri, Rwa Assi, Aisha Alraeesi, Hour Mohamed, Fakhraddin Alwajih, Abdelrahman Mohamed, Abdellah El Mekki, El Moatez Billah Nagoudi, Benelhadj Djelloul Mama Saadia, Hamzah A. Alsayadi, Walid Al-Dhabyani, and 8 others. 2024. [Casablanca: Data and models for multidialectal Arabic speech recognition](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 21745–21758, Miami, Florida, USA. Association for Computational Linguistics.
- Bashar Talafha, Hawau Olamide Toyin, Peter Sullivan, AbdelRahim A. Elmadany, Abdurrahman Juma, Amirbek Djanibekov, Chiyu Zhang, Hamad Alshehhi, Hanan Aldarmaki, Mustafa Jarrar, Nizar Habash, and Muhammad Abdul-Mageed. 2025. [NADI 2025: The first multidialectal Arabic speech processing shared task](#). In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 720–733, Suzhou, China. Association for Computational Linguistics.
- Jörgen Valk and Tanel Alumäe. 2021. [VoxLingua107: A dataset for spoken language recognition](#). In *IEEE Spoken Language Technology Workshop (SLT)*.
- Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, and 3 others. 2020. [Transformers: State-of-the-art natural language processing](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*.
- Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. 2021. [mT5: A massively multilingual pre-trained text-to-text transformer](#). In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics.
- YatharthS. 2025. [LavaSR: Speech enhancement and bandwidth extension](#). Hugging Face model.
- Mario Zusag, Laurin Wagner, and Bernhard Thallinger. 2024. [CrisperWhisper: Accurate timestamps on verbatim speech transcriptions](#). *arXiv preprint arXiv:2408.16589*.

A Hyperparameters and Tooling

Tooling: PyTorch 2.11 (<https://pytorch.org>), Transformers 5.13 (<https://github.com/huggingface/transformers>), bitsandbytes 0.49 (<https://github.com/bitsandbytes-foundation/bitsandbytes>), jiwer 4.0 (<https://github.com/jitsi/jiwer>), LavaSR (<https://huggingface.co/YatharthS/LavaSR>). The silver-label experiments and the 1.2 label audit used Gemini 3.5 Flash (Google DeepMind, 2026).

B Negative Results

Table 4 catalogs the interventions that tied or regressed. Systems referenced there: CrisperWhisper (Zusag et al., 2024), AraT5 (Nagoudi et al., 2022), Qwen3-ASR (Qwen Team, 2026a), and Gemini 3.5

Setting	Value
ASR base	Whisper large-v3 (1.5B)
Precision	bf16 + grad. checkpointing
In-task FT lr	1×10^{-5}
Continue-FT lr	5×10^{-6} , cosine
Optimizer	paged AdamW 8-bit
Batch	$4 \times$ accum. 4 (eff. 16–32)
SpecAugment (1.3)	$p_{\text{time}}=p_{\text{feat}}=0.05$
Checkpoint rule	best on in-domain holdout
Decoding	greedy; chunked 30 s/5 s stride
De-looping	collapse 1–4-gram repeats >2
Correctors	mT5-large, fp32, beam 4
DID training	frozen-head then full; class-weighted CE
DID fusion	0.5/0.5 raw/LavaSR + $w \cdot \text{text}$
Software	PyTorch 2.11, Transformers 5.13, bitsandbytes 0.49, jiwer 4.0
Hardware	1×32 GB GPU

Table 3: Shared hyperparameters across systems.

Flash (Google DeepMind, 2026). Two audits deserve note: (i) an audio-hash audit (MD5 over raw bytes and decoded PCM) found *zero* overlap between the 1.3 blind set and the public Tunisian corpora, so training on public test splits was never going to leak signal; (ii) a full Gemini re-transcription of the 1.2 dev set found zero clips whose gold is incoherent while Gemini’s transcript is coherent (median CER of Gemini hypotheses against the gold references: 0.150), ruling out label noise as the cause of the 0.48 plateau.

C Data Inventory

Table 5 lists the principal training resources with sizes; all are public.

D Per-Dialect Results

Subtask 1.1. The per-dialect spread is large and stable across systems: Palestine, Jordan, and UAE are easiest (≈ 0.41 – 0.44 WER for the Stage-2 model); Yemen, Egypt, and Morocco intermediate; and the Maghrebi/Saharan tail hardest — Algeria improved $0.71 \rightarrow 0.662$ with targeted conversational data, while Mauritania stayed at 0.78 – 0.79 under every intervention we tried, making it the dominant open problem for this dialect set.

Subtask 3. Table 6 gives the official per-dialect automatic evaluation. Quality prediction and intelligibility diverge: Morocco and Sudan score highest on UTMOS yet Morocco has the worst WER, while Saudi and Palestine pair mid UTMOS with the best WER; Yemen is weakest on every metric, consistent with its reference being auto-transcribed.

Intervention	Outcome (blind)
<i>Subtask 1.1</i>	
Gemini silver labels	regressed dev + blind
New Algerian YT data	dev win, blind $0.52 \rightarrow 0.53$
Per-dialect routing ($\times 3$)	0.52 vs. 0.5111
Cohere ASR (LoRA FT)	val 0.665 vs. ours 0.325
CrisperWhisper base	holdout 0.519 vs. 0.402
Test-side enhancement	$0.52 \rightarrow 0.55$ – 0.88
<i>Subtask 1.2</i>	
SADA-Khaliji gold	holdout win, blind tie
mT5 / Qwen correctors	tie (0.48)
Qwen3-ASR native FT	0.59
Gemini label audit	0 broken-gold clips
<i>Subtask 1.3</i>	
3rd/4th epoch; warm cont.	15.4 / overfit
Egyptian–English CS data	15.42
“Cleaner” re-processing	val 19.44 vs. 17.26
Beam/n-gram-block decode	tie to +4pt val
AraT5 / mT5 / Qwen corr.	+1.4 / tie / +5.9
Public-test-split training	no transfer (0 hash overlap)
<i>Task 2</i>	
7 alt. backbones	0.05 – 0.33
Higher-capacity head	$0.48 \rightarrow 0.38$
160K cross-source data	0.22
Extra TTA/enhanced views	dilute to 0.44 – 0.49
Corrected transcripts	0.509 vs. 0.5114

Table 4: Negative results across subtasks; each row ties or regresses relative to the submitted system.

Resource	Clips
NADI 1.1 official train	23,608
Casablanca (gold, 8 dialects)	13,637
Mixed-ASR dev (only labeled 1.2 data)	3,152
NADI_TUN train	~ 24.9 K
tunisian-asr-cleaned (external gold)	~ 46 K
ADI-20 curated subset	67K
lahgtna (base mix, 267 h)	54,600
CV18 + FLEURS + MASC (base mix)	—
SADA22-Khaliji (probe; no blind gain)	6K

Table 5: Principal training resources (KAND CA, 2026a; Talafha et al., 2024; Bougares, 2026; FARUKx-AUTO, 2026; Shon et al., 2020; Ardila et al., 2020; Conneau et al., 2023; Al-Fetyani et al., 2021; SDAIA, 2022).

Dial.	N	UTMOS	N-MOS	N-TTS	WER	CER
ae	537	2.73	3.71	3.45	0.313	0.155
dz	293	2.60	3.81	2.96	0.446	0.130
eg	424	2.81	2.75	3.75	0.205	0.054
jo	631	2.60	4.21	3.18	0.328	0.082
ma	201	3.23	3.52	3.05	0.555	0.216
ps	750	2.58	4.23	3.09	0.202	0.057
sa	100	2.79	3.85	3.45	0.143	0.049
sd	114	3.19	4.46	3.98	0.357	0.100
tn	623	3.04	2.74	3.20	0.445	0.122
ye	328	2.17	2.65	2.46	0.479	0.251
all	4,001	2.72	3.57	3.21	0.325	0.114

Table 6: Official Subtask 3 automatic evaluation per dialect: UTMOS (Saeki et al., 2022), NISQA-MOS and NISQA-TTS (Mittag et al., 2021), and WER/CER computed by the organizers with Omnilingual ASR 7B (Meta AI, 2025).